PRINCE TAWIAH

Field and Facilities Engineering | Critical Power Systems | Site Deployment
Houghton, MI | (906) 319-8851 | ptawia3@mtu.edu | ptawiah.st@gmail.com | linkedin.com/in/ptawia3

PROFESSIONAL SUMMARY

Hands-on engineer and technician with 5+ years (2019–2025) maintaining and troubleshooting mission-critical facility equipment in 24/7 operations, including electrical panels, power distribution, uninterruptible power supply (UPS) systems, backup generators, and automatic transfer switch (ATS) switchgear. Experienced in routine inspections, generator and ATS transfer testing, preventive and corrective maintenance, installation and commissioning support, lockout/tagout (LOTO) safety, technical documentation in Jira, and leading a 10-person technical team. B.Sc. in Telecommunication Engineering; M.S. in Data Science (expected Dec 2026) with research on data center IT power demand. Willing to travel, relocate, and work rotating shifts.

EXPERIENCE

Research Assistant, Michigan Technological University **Feb 2026 – Present**

- Model AI workload power draw from GPU and CPU utilization to estimate IT power demand in large-scale data centers

Maintenance Technician, Zipline International Inc. **Dec 2019 – Jan 2025**

24/7 critical-operations distribution center: Mpanya, Ashanti Mampong, Ghana

- Performed routine inspections and preventive and corrective maintenance on critical site power, including electrical panels, power distribution components, and UPS systems, to ensure continuous power availability
- Performed generator readiness and ATS transfer testing, transferring site load from utility to generator power and back while monitoring voltage, frequency, and engine readings; serviced generators and checked fuel systems
- Supported installation, testing, and commissioning of electrical equipment during site upgrades; routed, dressed, and labeled power and control cables to separation and bend-radius standards
- Followed strict LOTO and electrical safety procedures when working around energized systems and critical infrastructure
- Maintained and troubleshot UAV battery charging systems to keep launch operations available
- Participated in incident investigations and process reviews; helped close a standard operating procedure (SOP) gap by adding a mandatory engineering clearance step before aircraft return to service
- Diagnosed and repaired electrical and mechanical UAV systems, sustaining >98% monthly fleet uptime with 0% maintenance-related aircraft retirements
- Supervised a team of 10 flight technicians as Maintenance Lead; documented and tracked work in Jira and built triage and escalation workflows that accelerated resolution
- Built maintenance and inventory key performance indicator (KPI) dashboards (6,000+ views) and operations dashboards used in 7 countries (10,000+ views)

Systems Installation Lead (New Site Deployments), Zipline International Inc., Nigeria **Jan 2022 – Oct 2022**

- Relocated to Nigeria to deploy and bring up multiple new distribution centers, from installation through operational readiness
- Led installation of UAV battery charging systems at each new site
- Supported site power systems installation and coordinated cross-functional teams through go-live

Interim Operations Manager, Zipline International Inc. **Jan 2022 – Dec 2024**

- Managed a team of 10 flight technicians during leadership gaps; trained staff and audited field work for compliance with safety and operating procedures

Intern Engineer, Ghana Grid Company Ltd., Tema **Jul 2017 – Aug 2017**

- Traced and tested fiber optic cable runs with an optical time-domain reflectometer (OTDR) to locate faults and verify link integrity
- Monitored network performance and compiled statistical reports on link status for the engineering team

EDUCATION

M.S. Data Science, Michigan Technological University **Expected Dec 2026**

B.Sc. Telecommunication Engineering, Kwame Nkrumah University of Science and Technology **Jun 2018**

SKILLS

Core Skills: Electrical Systems (Panels, Power Distribution) | Generators, UPS and ATS Switchgear | Facility Inspections | Preventive and Corrective Maintenance | Installation, Testing and Commissioning | LOTO and Electrical Safety (OSHA 10) | Technical Documentation and Ticketing (Jira) | Low-Voltage and Fiber Cabling (OTDR) | Root-Cause Analysis

Software and Tools: Jira, Microsoft Office (Excel), Google Workspace, Python, SQL, Snowflake, VMware vCenter, Linux

CERTIFICATIONS AND AWARDS

OSHA 10-Hour Construction | Best Flight Operator, Zipline Ghana (2023) | Customer Obsession and Servant Leadership Award, Zipline Ghana (2022)